2 SEMICONDUCTOR CRYSTAL STRUCTURE

2.1 Types of Solids

Solids are characterised by the size of an ordered region within the material. The ordered region is a spatial volume in which atoms or molecules have a regular geometric arrangement, or periodicity.

Three general types of solids (see Fig. 2.1)

- Amorphous materials
  - Ordered region is only within a few atomic or molecular dimensions (a few to tens of angstroms; 1Å = 10^{-10} m)
- Polycrystalline materials
  - High degree of order over many atomic or molecular dimensions

- Ordered regions (or single crystal regions) vary in size and orientation with respect to one another
- These single crystal regions are called grains and are separated from one another by grain boundaries

- Single crystal
  - High degree or order, or regular geometric periodicity, throughout entire volume of material
  - Electrical properties are superior to those of non-single crystal materials (this is because grain boundaries tend to degrade electrical characteristics)

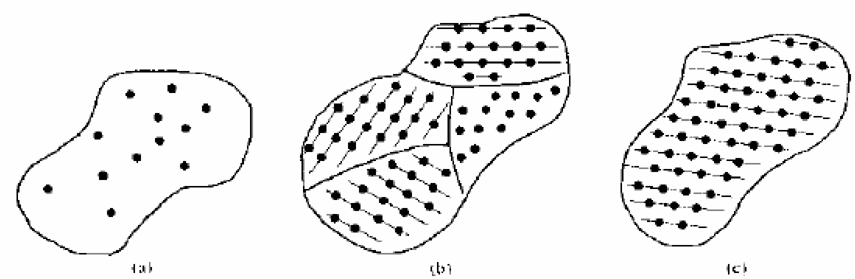


Fig.2.1 Schematics of three general types of crystals: (a) amorphous, (b) polycrystalline, (c) single crystal

2.2 Space Lattice, Primitive and Unit Cell

Consider a single crystal

- Periodic arrangement of points in a space is called a lattice. The point is called a lattice point. Fig. 2.2 shows an infinite two-dimensional (2-D) array of lattice points.
  - Each lattice point can be translated a distance a_1 and a distance b_1 in a second non-colinear direction to generate the 2-D lattice.
  - A third non-colinear translation will produce the three dimensional (3-D) lattice. The lattice directions need not be perpendicular

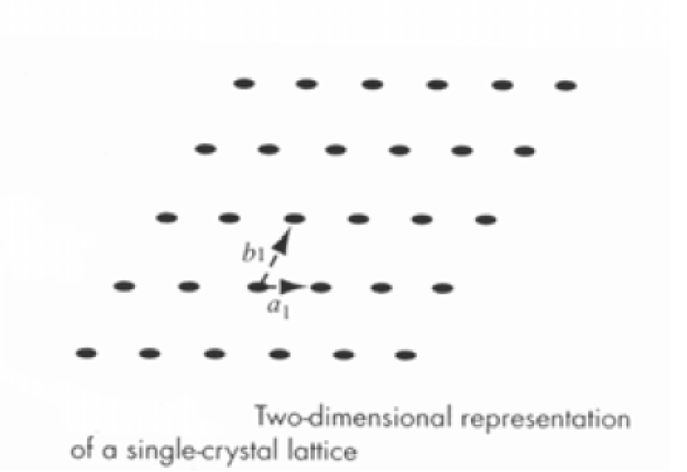


Fig. 2.2

- A crystal structure is formed when an atom, group of atoms/molecules is attached identically to each lattice point

When considering a lattice,

- Do not need to consider the entire lattice. Need to look only at a fundamental unit that is being repeated

- A unit cell is a small volume of crystal that can be used to reproduce the entire crystal. It is a representative of the entire lattice. We can generate the entire lattice by repeating the unit cell throughout the crystal

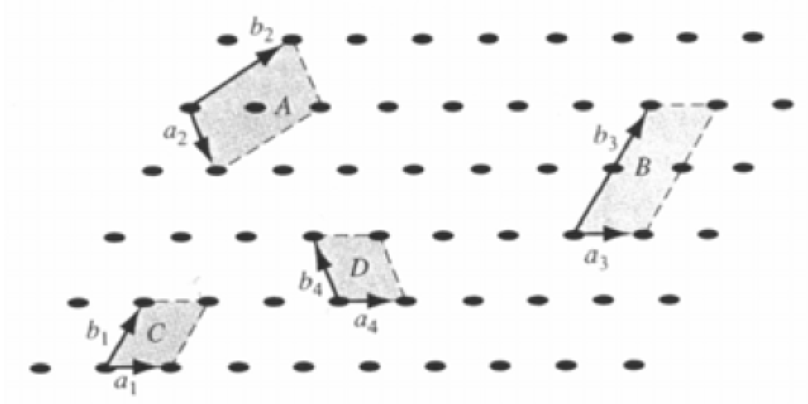
    - A unit cell is not unique entity

    - Several possible unit cell in 2-D lattice is shown in Fig. 2.3

- A primitive cell is the smallest unit cell that can be repeated to form lattice

- In many cases, more convenient to use a unit cell

  - They can be chosen to have orthogonal sides, whereas sides of primitive cell may not be orthogonal as shown in Fig. 2.4.



Two-dimensional representation of a single-crystal lattice showing various possible unit cells

Fig. 2.3

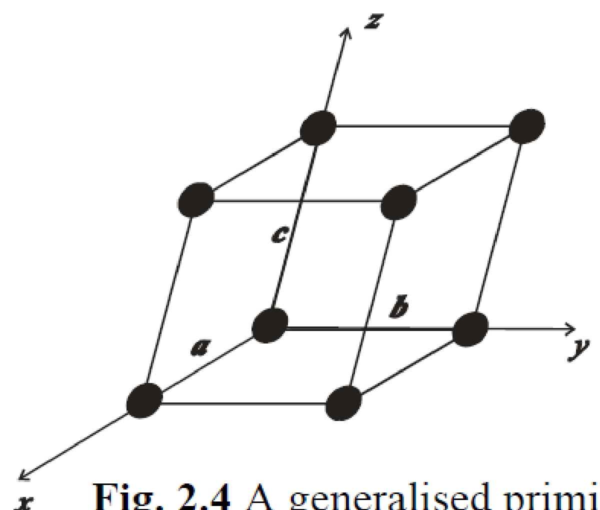
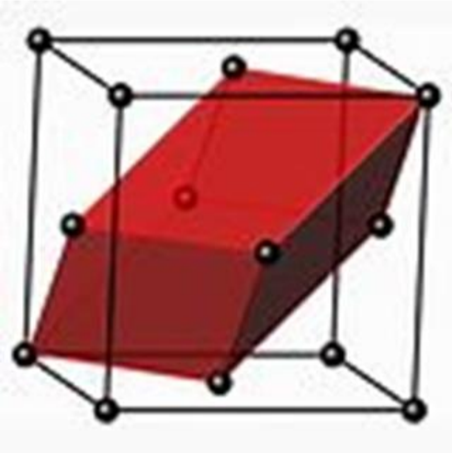


Fig. 2.4 A generalised primitive unit cell

Unit Cell Vs Primitive Cells



A unit cell is the smallest group of atoms that has the overall symmetry of a crystal, and from which the entire lattice can be built up by repetition in three dimensions. Therefore, unit cells are the repeating units of crystal lattices.

A primitive cell is the smallest possible unit cell of a lattice, having lattice points only at each of its eight vertices. Hence, it is the simplest form of unit cells.

2.3 Crystallographic directions and planes

Consider a unit cell defined by 3 vectors ***a*, *b*** and ***c***. The dimensions a, b and c are unit cell dimensions or lattice constant. For cubic unit cell the dimensions $a = b = c$.

A crystallographic direction is defined as a line between two points, a vector. The following steps can be employed to determine the crystallographic directions

- determine the origin

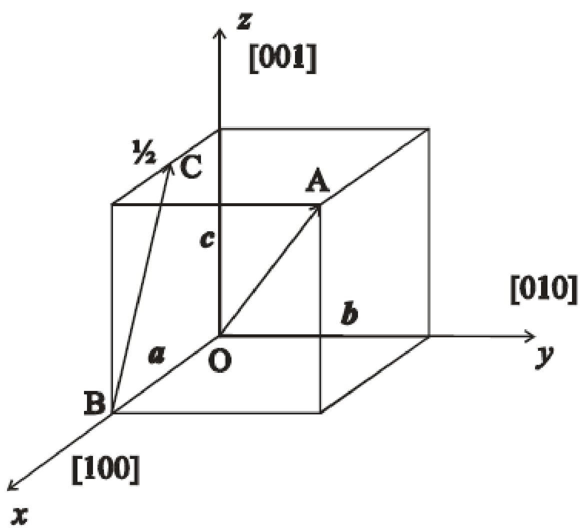
- determine the initial and the final positions of the vector in terms of the lattice constant a, b and c

- determine the vector: (final position) – (initial position)

- write down the vector in terms of the numbers of the lattice constant
- multiply or divide these numbers by a common factor to reduce them to the smallest integer values
- the three indices are enclosed in square brackets [*uvw*]. The *u*, *v* and *w* integers correspond to the reduced projections along *x*, *y* and *z*
- negative indices are represented by a bar over the appropriate index

Some examples (Example 2.1) are given to determine the crystallographic direction using the above procedures.

Examples 2.1



Vector OA

	x	y	z
Final position	$1a$	$1b$	$1c$
Initial position	$0a$	$0b$	$0c$
Vector OA	$1a$	$1b$	$1c$
Vector OA (numbers only)	1	1	1
Reduction	1	1	1
Enclosure	$[111]$		

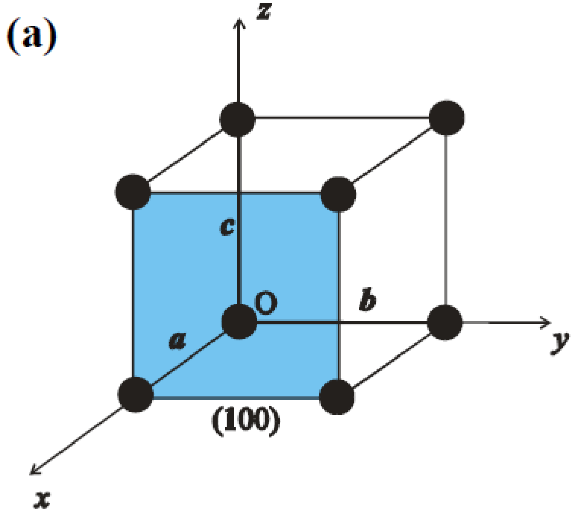
Vector BC

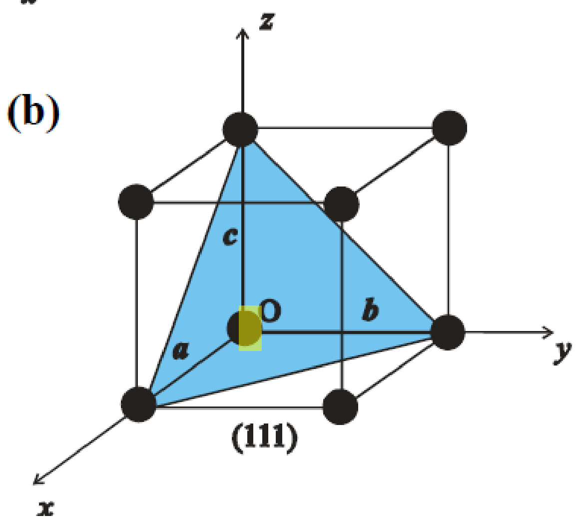
	x	y	z
Final position	$1/2a$	$0b$	$1c$
Initial position	$1a$	$0b$	$0c$
Vector OA	$-1/2a$	$0b$	$1c$
Vector OA (numbers only)	$-1/2$	0	1
Reduction	-1	0	2
Enclosure	$[\bar{1}02]$		

Planes in a crystal are specified by Miller indices (*hkl*). The following steps are employed to determine the Miller indices of the planes

- determine the origin, make sure that the planes do not pass through the origin
- determine the intercepts on the axes in terms of the lattice constant *a*, *b* and *c*
- write down the intercepts in terms of the numbers of the lattice constant
- take the reciprocal of these numbers
- multiply or divide the reciprocal to reduce them to usually smallest three integers
- the integers are enclosed in parentheses (*hkl*)
- again, negative indices are represented by a bar over the appropriate index

Example 2.2





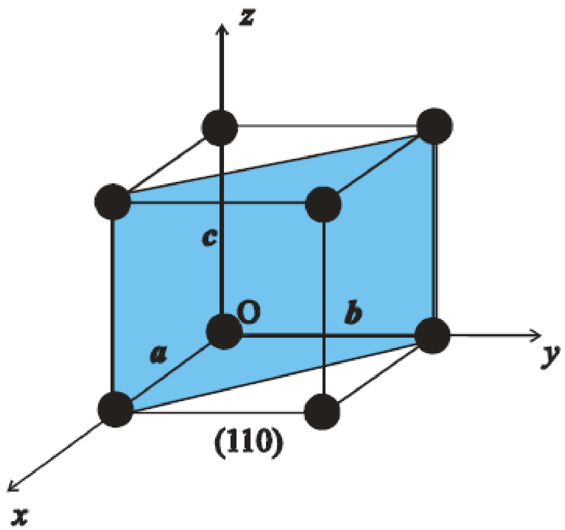
(100) plane (origin at O)

	x	y	z
Intercepts	$1a$	$\infty\ b$	$\infty\ c$
Intercepts (numbers only)	1	∞	∞
Reciprocal	1	0	0
Reduction	1	0	0
Enclosure	(100)		

(111) plane (origin at O)

	x	y	z
Intercepts	$1a$	$1\ b$	$1\ c$
Intercepts (numbers only)	1	1	1
Reciprocal	1	1	1
Reduction	1	1	1
Enclosure	(111)		

(c)



(110) plane (origin at O)

	x	y	z
Intercepts	$1a$	$1\ b$	$\infty\ c$
Intercepts (numbers only)	1	1	∞
Reciprocal	1	1	0
Reduction	1	1	0
Enclosure	(110)		

2.4 Crystal Structures

Generalised 3D unit cell

- The relationship between a unit cell and a lattice is characterized by 3 vectors, ***a***, ***b***, ***c***, which need not be perpendicular and may, or may not, be equal in length

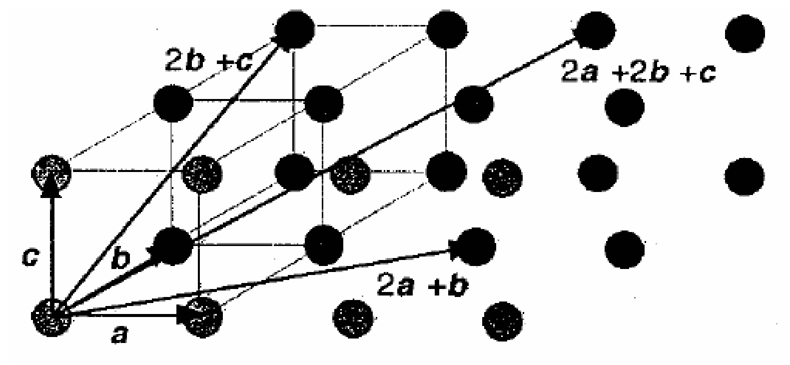


Fig. 2.5

- Every equivalent lattice point in 3D crystal can be found using

$$\boldsymbol{r} = p\boldsymbol{a} + q\boldsymbol{b} + s\boldsymbol{c} \qquad \textbf{(2.1)}$$

where p, q and s are integer (see Fig. 2.5)

- Note that location of the origin is arbitrary

2.4.1 Basic crystal structures

Three simple structures are

- Simple cubic (sc)
- Body-centered cubic (bcc)
- Face-centered cubic (fcc)

The vectors ***a*, *b*** and ***c*** of the unit cells are perpendicular to each other and have equal lengths.

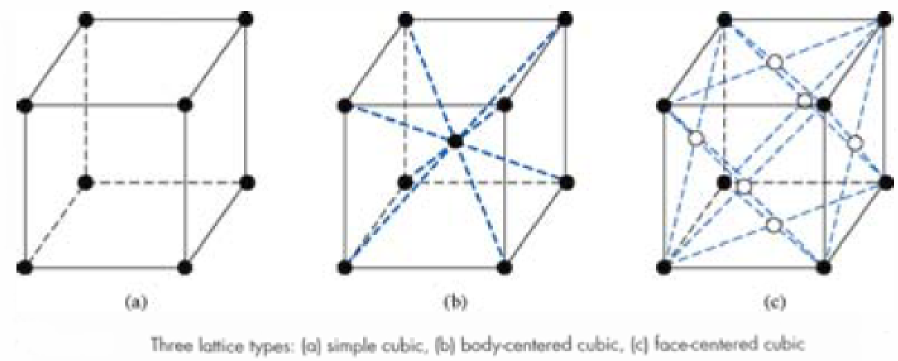


Fig. 2.6

The simple cubic, body-centered cubic and face-centered cubic structure are shown in Fig. 2.6.

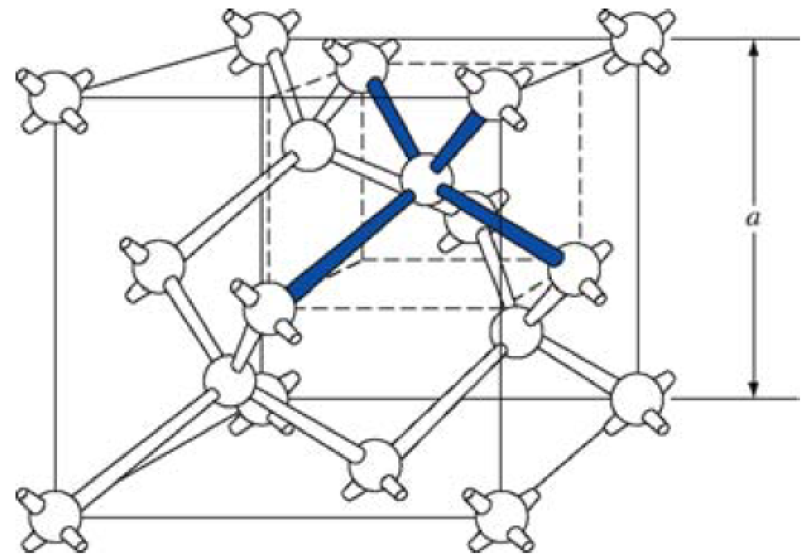
- for simple cubic crystal structure
    - one lattice point located at each corner of the cube
- for body-centered cubic crystal structure
    - one lattice point located at each corner of the cube and an additional lattice point at the center of the cube
- for face-centered cubic crystal structure
    - one lattice point located at each corner of the cube and additional lattice points on each face plane
- To have a crystal, one or more atoms/molecules can be placed at each lattice point
- Length of the cube side is termed as the *lattice constant*

- By knowing the crystal structure and the lattice dimensions, we are able to determine the surface density (# atoms/cm^2) and volume density of atoms (# atoms/cm^3)

2.5 Semiconductor Crystals

Fig. 2.7 shows diamond crystal structure

- Silicon and Germanium have this structure
- All atoms are of the same species
- Any atom within the diamond structure will have four nearest neighboring atoms
  - they are bonded by covalent bonding



The diamond structure

Fig. 2.7

- Silicon crystal can be regarded as a face-centered cubic structure with 2 silicon (Si) atoms at each lattice point
    - the relative position of the Si atoms at each lattice point are (0,0,0) and (a/4,a/4,a/4)

This is shown in Fig. 2.8

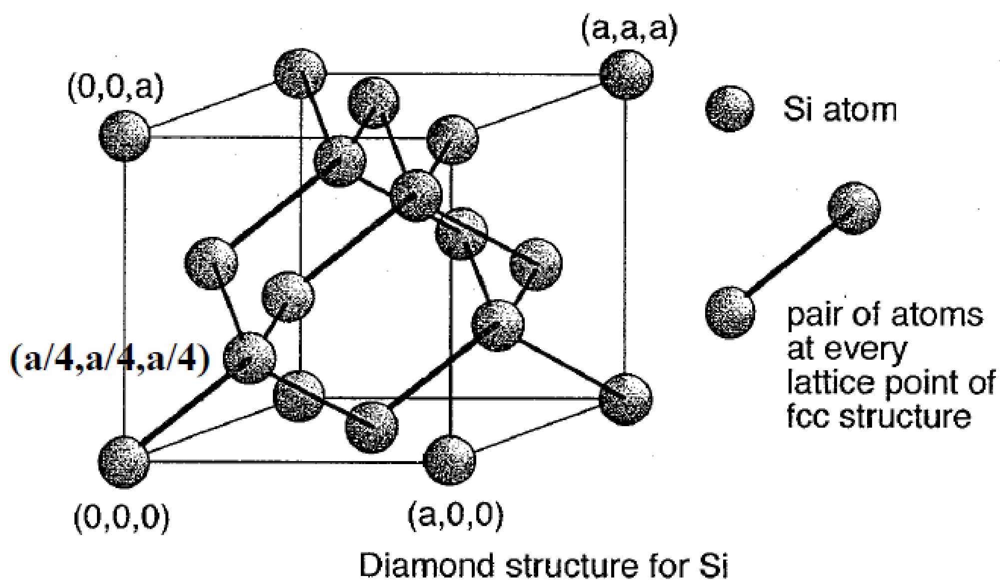


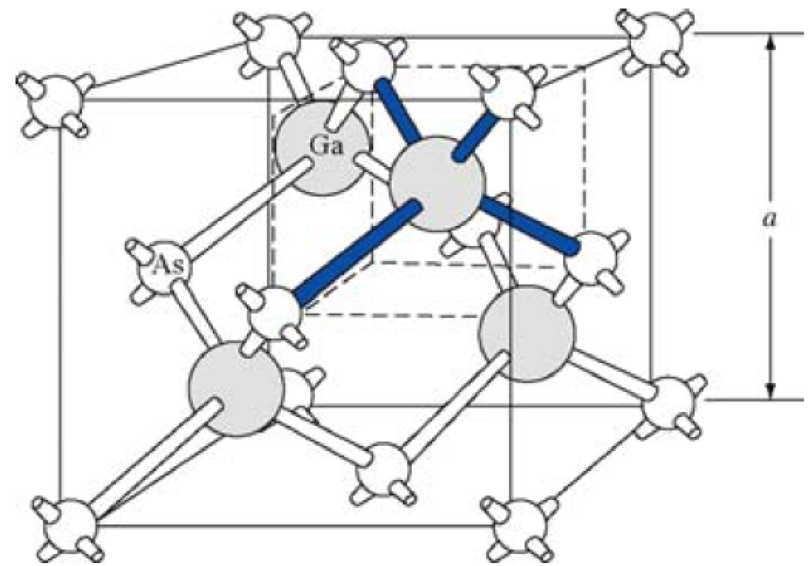
Fig. 2.8

Fig. 2.9 shows a zincblende (sphalerite) crystal structure

- This structure differs from the diamond structure only in that there are two different types of atoms in the lattice

- Most compound semiconductors, e.g. GaAs (gallium arsenide) and InP (indium phosphide) have this structure

- For a GaAs crystal, each Ga atom has 4 nearest As neighbors and each As atom has 4 nearest Ga neighbors
    - they are bonded by covalent bonding



The zincblende (sphalerite) lattice of GaAs

Fig. 2.9

- GaAs crystals can also be considered as face-centered cubic with one Ga atom and one As atom at each lattice point
    - the relative position of the Ga and As atoms at each lattice point are (0,0,0) and (a/4,a/4,a/4)

This is shown in Fig. 2.10

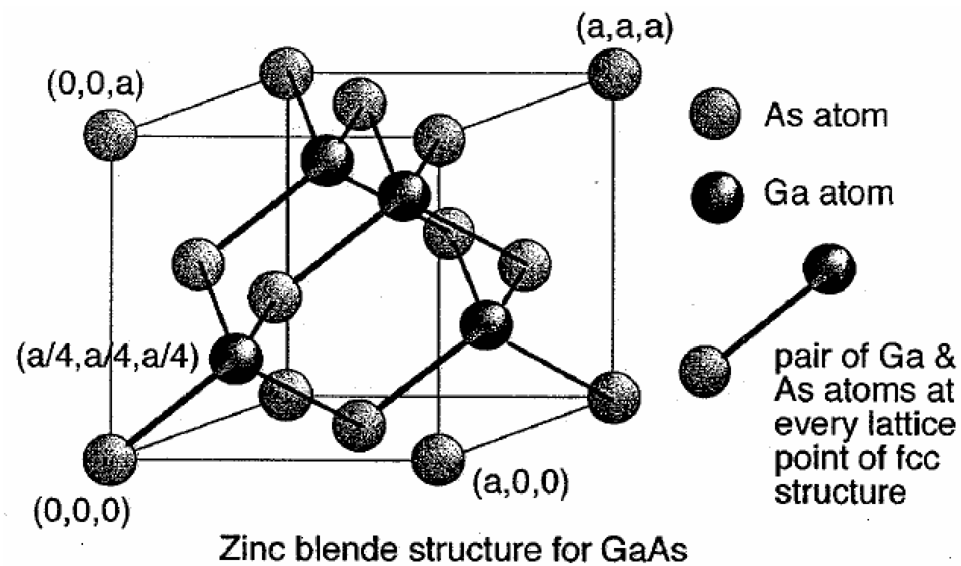
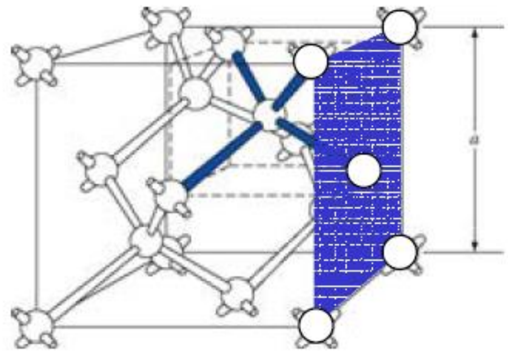


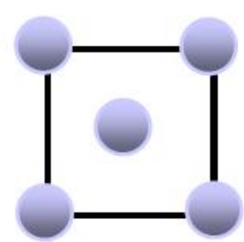
Fig. 2.10

Example 2.3

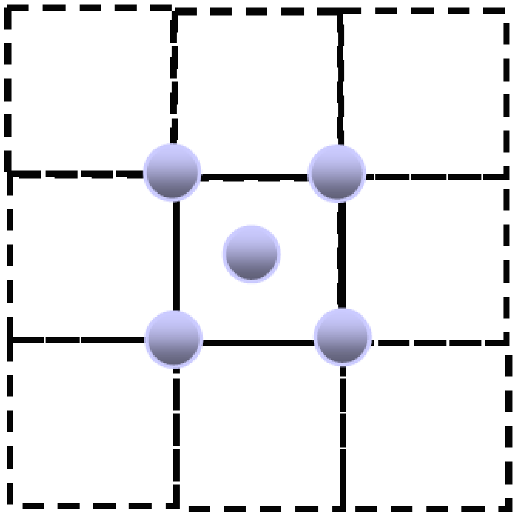
Calculate the surface density of atoms (# atoms/cm^2) on the shaded plane (010 plane) of a silicon crystal shown. The lattice constant for silicon is 5.43Å (1Å = 10^{-10} m).



The silicon crystal has the face-centered cubic structure. The indicated plane can be redrawn as below



Notice that the atom at each corner is shared by 4 similar equivalent lattice planes, so each corner atom effectively contributes 1/4 atom to the indicated lattice plane.



The atom in the middle of the plane is not shared by any other plane.

Hence, no. of atoms on the plane is $4(1/4) + 1 = 2$ atoms

Therefore,

$$\text{Surface density} = \frac{\text{No. of atoms}}{a^2} = \frac{2}{\left(5.43\times10^{-8}\ \text{cm}\right)^2}$$

$$= 6.78\times10^{14}\ \ \text{atoms/cm}^2$$

Calculate the volume density (# atoms /cm^3) in one unit cell of silicon crystal.

Total number of atoms in one unit cell = 8 x 1/8 + 6 x 1/2 + 4 = 1+ 3+ 4 = 8

Volume density = 8 / a^3